Lab 5: Static Code Analysis

Movie Theater Seat Reservation System (theater_system.py)

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GitHub repo: https://github.com/Priyanka0205-CSE/L5_09_movie_seats

Summary

Tool	Before (original code)	After (fixed code)
Pylint	5.87 / 10 (19 messages)	10.00 / 10
Bandit	3 issues (2 High, 1 Low)	No issues identified
Flake8	11 issues	No issues (empty output)

Part 1: BEFORE (original code)

1.1 Pylint

Command: `pylint theater_system.py`

```
************* Module theater_system
theater_system.py:40:0: C0301: Line too long (104/100) (line-too-long)
theater_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
theater_system.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:9:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
theater_system.py:12:27: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
theater_system.py:15:23: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
theater_system.py:18:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:21:4: W0702: No exception type(s) specified (bare-except)
theater_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:28:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
theater_system.py:28:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
theater_system.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:32:4: W0603: Using the global statement (global-statement)
theater_system.py:33:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
theater_system.py:33:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
theater_system.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)
theater_system.py:37:0: C0103: Function name "printSeatingSummary" doesn't conform to snake_case naming style
    (invalid-name)
theater_system.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 5.87/10
```

1.2 Bandit

Command: `bandit -r theater_system.py`

```
Run started:2026-10-22 15:12:30.073686+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\theater_system.py:21:4
20          shows[show_id]["booked"].remove(seat_number)
21      except:
22          pass
23

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\theater_system.py:25:11
24  def ticket_code(show_id, seat_number):
25      return hashlib.md5((show_id + str(seat_number)).encode()).hexdigest()[:8]
26

--------------------------------------------------
>> Issue: [B103:set_bad_file_permissions] Chmod setting a permissive mask 0o777 on file (seats.json).
   Severity: High   Confidence: High
   CWE: CWE-732 (https://cwe.mitre.org/data/definitions/732.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b103_set_bad_file_permissions.html
   Location: .\theater_system.py:55:4
54      printSeatingSummary()
55      os.chmod("seats.json", 0o777)
56

--------------------------------------------------

Code scanned:
    Total lines of code: 47
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 1
```

```
        Medium: 0
        High: 2
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 3
Files skipped (0):
```

1.3 Flake8

Command: `flake8 theater_system.py`

```
theater_system.py:9:1: E302 expected 2 blank lines, found 1
theater_system.py:12:80: E501 line too long (83 > 79 characters)
theater_system.py:18:1: E302 expected 2 blank lines, found 1
theater_system.py:21:5: E722 do not use bare 'except'
theater_system.py:24:1: E302 expected 2 blank lines, found 1
theater_system.py:27:1: E302 expected 2 blank lines, found 1
theater_system.py:31:1: E302 expected 2 blank lines, found 1
theater_system.py:37:1: E302 expected 2 blank lines, found 1
theater_system.py:40:80: E501 line too long (104 > 79 characters)
theater_system.py:42:1: E302 expected 2 blank lines, found 1
theater_system.py:57:1: E305 expected 2 blank lines after class or function definition, found 1
```

Part 2: AFTER (fixed code)

2.1 Pylint

Command: `pylint theater_system.py`

```
--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 9.85/10, +0.15)
```

2.2 Bandit

Command: `bandit -r theater_system.py`

```
Run started:2026-10-22 15:37:23.256346+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 82
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
```

2.3 Flake8

Command: `flake8 theater_system.py`

```
(no output - no issues found)
```